

15.7

15 在交变磁场中,铁心中的磁滞损耗和涡流损耗是怎样产生的?它们与哪些因素有关?

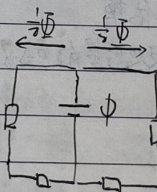
磁滞损耗,铁磁材料置于交变磁场中,材料被反复交变磁化,磁畴相互不停地摩擦而消耗能量,并以热量的形式表现出来,造成的损耗。

涡流损耗,铁心是导电的,当通过铁心的磁通随时间变化时,铁心中将产生感应电动势并引起环流,这些环流在铁心内部引起涡流损耗。

磁滞损耗,磁场的交变的频率,铁心的体积,磁滞回线的面积。

涡流损耗系数,其大小钢材厚度,频率(磁场的交变),铁心的体积,磁通密度的最大值。

$$17 \text{ 中 } A_0 = (1.15 \times 10^{-2} + 0.025 \times 10^{-2}) \times (0.025 \times 0.93 + 0.025 \times 10^{-2}) = 3 \times 10^{-4} \text{ m}^2$$



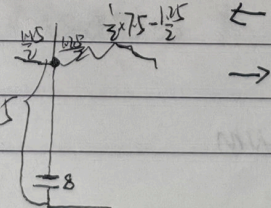
$$B_0 = \frac{\Phi}{A_0} = \frac{7.5 \times 10^{-4}}{2.5 \times 10^{-4}} = 1.25 \text{ T}$$

$$F = H_0 L_0 = \frac{B_0}{\mu_0} L_0 = \frac{1.25}{4\pi \times 10^{-7}} \times 0.025 \times 2 = 5000 = N_2 \cdot i = 0.5 \text{ A}$$

又考虑铁心的磁位降, $B = \frac{\Phi}{A_1} = \frac{7.5 \times 10^{-4}}{2.5 \times 10^{-2} \times 2.5 \times 10^{-2} \times 0.93} = 1.29 \text{ T}$

则 $H = 7 \times 10^2 \text{ A/m}$

Φ $l_2 = \frac{1}{2} \times 7.5 \times 10^{-2} \times 2 + 5 \times 10^{-2} \times 2 - 1.15 \times \frac{1}{2} \times 10^{-2} \times 2 = 0.1625 \text{ m}$



$$F = H L + H_0 l_2 = 0.1625 \times 7 \times 10^2 + 1000 = 613.8 \text{ A}$$

$$I = \frac{F}{N} = 0.614 \text{ A}$$

2-10, 15, 18, 21, 23, 24

10. 电枢电路和励磁电路的串联电阻，取决于输入电压，电枢电动势，反电势

4

$$15. \phi \quad P_i = U_N I_N = 17.6 \text{ kW}$$

$$\text{或 } P_i \eta_N = 14.9 \text{ kW}$$

$$3. P_i(1-\eta_N) = 2.64 \text{ kW}$$

$$4. I_f = \frac{U_N}{R_f} = 2.48 \text{ A}, I_a = 80 - 2.48 = 77.5 \text{ A}$$

$$P_{\text{out}} = I_a^2 R_a = 77.5^2 \times 0.08 = 480 \text{ W}$$

$$5. P_{\text{out}} = I_f^2 R_f = 14.6 \text{ W}$$

$$6. P_c = 20 \text{ W}, I_a = 15 \text{ A} \quad \text{铜耗+铁耗+机械损耗+附加损耗}$$

$$7. P_{\Delta} = 1\% P_N = 1\% \times 14.9 \text{ kW} = 150 \text{ W} \quad \text{按额定容量的1\%估算}$$

$$8. P_{\text{mech}} + P_{\text{fe}} = P - P_{\text{cu}} - P_{\text{at}} - P_c - P_{\Delta} \quad \text{有补偿绕组的，额定容量的0.5\%}$$

$$= 2640 \text{ W} - 480 - 146 - 150 - 150 = 1509 \text{ W}$$

$$2-18 \quad P_N = C_e \phi n_N = E_a \quad P_N = E_a \cdot I_a$$

$$I_a C_e \phi = \frac{17 \text{ kW}}{3000 \text{ r/min}}$$

$$T_e = \frac{P_N}{\omega} = C_T \phi I_a$$

$$\text{或 } 9.55 C_T \phi I_a = \frac{17 \text{ kW}}{3000} = 5.67 \text{ N}\cdot\text{m}$$

$$I_f = \frac{U_N}{R_f} = \frac{220}{187.5} = 1.18 \text{ A}$$

$$I_a = I_N - I_f = 88.9 \text{ A} - 1.18 \text{ A} = 87.7 \text{ A}$$

$$C_e \phi n_N = U_N - I_a R_a \approx 220 - 87.7 \times 0.08 = 213 \text{ V} \quad T_e = 9.55 C_T \phi I_a = 58.6 \text{ N}\cdot\text{m}$$

$$13. \eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{P_N}{U_N I_N} = \frac{17 \times 10^3}{220 \times 88.9} \times 100\% = 86.9\%$$

$$4. n_0 = U_N / C_e \phi = \frac{220}{0.07} \text{ r/min} = 3143 \text{ r/min}$$

$$15. R_g = 0.15 \Omega$$

$$n = \frac{U_N - I_a(R_a + R_g)}{C_e \phi} = \frac{220 - 87.7 \times (0.08 + 0.15)}{0.07} \text{ r/min} = 2812 \text{ r/min}$$

$$2-21 \text{ 并励 } 220 \text{ V}, R_a = 0.316 \Omega, I_{a0} = 2.8 \text{ A}, n = 1600 \text{ r/min}$$

1. 负载电流 52 A, $n \rightarrow 800 \text{ r/min}$ 串入多大的电阻 (电枢回路)

$$I_f = \frac{U_N}{R_f} = \frac{220}{187.5} = 1.18 \text{ A}$$

$$2-22 \quad P_N = 75 \text{ kW}, U_N = 110 \text{ V}, I_N = 82 \text{ A}, n_N = 1500 \text{ r/min}$$

电枢回路的总电阻 $R_a = 0.1014 \Omega, R_f = 46.7 \Omega$, 忽略电枢反作用

1. 求电动机 $I_a = 60 \text{ A}$ 的转速

$$I_f = \frac{U_N}{R_f} = \frac{110}{46.7} = 2.36 \text{ A}$$

$$I_a + I_f = I_N = 82 \text{ A} \quad I_a = 82 \text{ A} - 2.36 \text{ A} = 79.64 \text{ A}$$

$$E_a = U_N - I_a R_a = 110 - 79.64 \times 0.1014 = 102 \text{ V} \quad \text{电枢电动势} = C_e \phi n$$

$$I_a = 60 \text{ A 时 } E_a = U_N - I_a R_a = 110 - 60 \times 0.1014 = 104 \text{ V} = C_e \phi n$$

$$\frac{104}{102} = \frac{n}{1500} \quad n = 1529 \text{ r/min}$$

$$2. \text{ 主磁路饱和 } 15\% \quad \Phi_0 = 0.85 \Phi, T_e = C_T \phi I_a = C_T \Phi_0 I_a'$$

$$I_a' = \frac{0.85}{0.85} I_a = \frac{79.64}{0.85} = 93.7 \text{ A}$$

$$E_a' = U_N - I_a' R_a = 110.5 \text{ V} = C_e \cdot 0.85 \Phi \cdot n', n' = 1739 \text{ r/min}$$

2-23. $U_N = 220V$, 电路回路总电阻 $R = 0.032\Omega$, $R_f = 25\Omega$

上升: $U_a = U_N$, $I_a = 350A$, $n = 78r/min$

下降: 负载转矩近似不变, 电压及励磁电流, $n = 300r/min$

$$E_a = U_N - I_a R_a = 220 - 350 \times 0.032 = 208.8V$$

$$n = 300r/min, E_a' = \frac{n'}{n} E_a, E_a' = -78.8V$$

$$R_{\Sigma} = \frac{U_N - E_a'}{I_a} = \frac{220 + 78.8}{350} = 0.854\Omega$$

$$R_{\Sigma} = 0.854\Omega - R_a = 0.854 - 0.032 = 0.822\Omega$$

2-24. $100kW$, $230V$, 并励力, $N = 1000$ 匝

励磁下, $I_f = 7A$, $I = 94A$, $U = 230V$, 平复励力, 通过调整串励绕组匝数,

改为平复励力, 增加多少匝串励绕组(每极)使满载时端电压空载电压相等。

设增加匝数 N_a , $I_a I_a = (I_{fN} - I_{f0}) N_f$

绕组有 N_a 匝

$$N_a = \frac{I_{fN} - I_{f0}}{I_a} N_f = \frac{(94 - 7) \times 1000}{100 \times 10 / 230} = 5.52 \approx 6$$

阅 2025.9.25

2 4 5 15 15 16

3-2 变压器的励磁电抗 X_m 的物理意义: 变压器的励磁电抗 X_m 是表征铁心磁化性能的一个综合参数, 由于铁心的磁化曲线是非线性的, 故 X_m 也不是一个常数, 随铁心饱和程度增加而减小的。

实际运行时一次侧端电压不变, 主磁通变化很小, 在这种情况下, 可以认为 X_m 不变化。从 $E = (R_m + jX_m) I_m$ 知, 变压器中希望 X_m 大些, 这样可以减小励磁电流 I_m 。若变压器改用空气心, 则磁路的磁导率会下降, X_m 与磁导率成正比, 可见 X_m 是减小的。 X_m 与次绕组的匝数平方成正比, 可见 X_m 是减小的。

X_m 与次绕组的匝数平方成正比, 当次绕组匝数增加时, 励磁电抗 X_m 是增加的。 X_m 与磁路饱和程度有关, 铁心截面积增加, 磁路饱和程度下降, 则 X_m 变大。

3-4. 1) 高压绕组则接到了 $220V$, 产生更高的交流电压 $U_N = 444fN\Phi_m$ 可知, 低压绕组匝数 N_2 减少, 磁通量要增加很多才能平衡端电压 U_N 磁

2) X_m , I_0 , p_{Fe}

$E_1 = -j444fN_1\Phi_m$, 则 E_1 幅值增大, 因为 E_1 增大又因磁路材料饱和, 励磁电流增加很多, 电流过大可能烧坏绕组铁心。

3) 无电压输出, 因电流很大而烧坏变压器线圈, 因为此时感应电动势为 0, 变压器(次侧) 电流 $I_2 = \frac{U_N}{Z_2}$

火频率提高 20% 由 $U_N = 444fN\Phi_m$ 可知, 磁通量 Φ_m 减少, 磁路的饱和程度减小, 则励磁电抗 X_m 增加, 空载电流是减小的。已知磁通量 Φ_m 用频率提高而减少, 则磁路的磁密 B 由减小铁耗 p_{Fe} 与磁密平方成正比, 相对而言, 铁耗 p_{Fe} 不是减小的?

3-5. $U_{IN}/U_{UN} = 220V/110V$, 空载电流 $I_{0I} = 2I_{0II}$

二次侧前空载电压是否相等
不相等

不计励磁阻抗, $Z_{mI} = \frac{1}{2}Z_{mII}$

串联后 $U_{I1} + U_{I2} = 440V$, 又根据阻抗分配得 $U_{I1} = \frac{1}{3} \times 440V = 146.7V$

$$U_{I2} = \frac{2}{3} \times 440V \approx 293.3V$$

由电压 $U_{I2} = 73.3V$, $U_{0I} = 146.6V$

3-13 $Z_k = \frac{U_k}{I_k} = \frac{157V}{7A} = 22.43\Omega$ $R_k = \frac{P_k}{I_k^2} = \frac{615}{7^2} = 12.55\Omega$
 $X_k = \sqrt{Z_k^2 - R_k^2} = 18.59\Omega$

空载铁耗 $P_{Fe} \approx P_0 = 24W$ 空载铜耗为0
满载铜耗 $P_{Cu} \approx P_k = 615W$ 满载铁耗为0

3 $I_N = \frac{S_N}{U_N} \approx 9.44A$

$\eta = \beta \frac{I_N R_k \cos \varphi_2 + I_N X_k \sin \varphi_2}{U_N} \approx 100\% = 1.87\%$

$U_2 = U_N (1 - \Delta U) \approx 471.0V$

$\gamma = \frac{P_2}{P_1} = (1 - \frac{P_0 + \beta P_k}{\beta S_N \cos \varphi_2 + P_0 + \beta P_k}) = 98.2\%$

3-15 采用“时钟表示法”, 把高低压绕组的两个线电压三角形的重心重合在一起, 并把高压侧线电压三角形的一条中线作为时钟的长针, 指向何向钟面12, 再把低压侧线电压三角形中相对应的中线作为时针短针, 它所指的钟点就是该变压器联结组的“标号”。

又三相 Yd 的三次谐波难以抵消, 谐波太大。

以 Yd 联结的三相变压器, 二次绕组存在三次谐波作用, 与 Yd 联结时在一次绕组中存在的三次谐波电流作用是相同的, 二次谐波电流会磁化, 改善电动势波形。

3-16

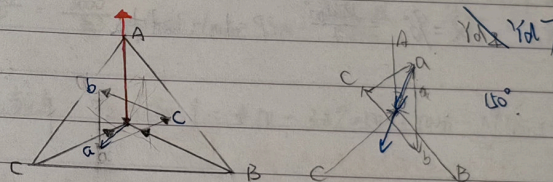


图 2025-10-29

4-16 若在三相对称三相绕组(三相绕组匝数相同,在空间相差 120° 电角度)中通入三相对称三相电流 $i_a = I_m \cos \omega t$, $i_b = I_m \sin \omega t$, 试用数学分析法证明合成磁动势为旋转磁动势。

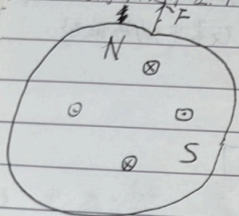
$$i_{a1} = I_{m1} \cos \frac{\pi}{6} \times \cos \omega t \quad i_{b1} = I_{m1} \sin \frac{\pi}{6} \times \sin \omega t$$

$$i_{a1} = I_{m1} \cos(\frac{\pi}{6} \times 180^\circ) \cos \omega t = I_{m1} \cos \omega t$$

$$i_{b1} = I_{m1} \cos(\frac{\pi}{6} \times 180^\circ - 180^\circ) \cos \omega t = -I_{m1} \cos \omega t$$

利用积化和差和差化积

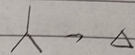
$$I_1 = i_{a1} + i_{b1} = I \cos(\omega t - \frac{\pi}{6} \times 180^\circ)$$



$$\alpha = \frac{P \times 90^\circ}{Q}$$

$$\alpha = 90^\circ = \frac{P \times 90^\circ}{4}, P=1, 1 \text{ 对极}$$

5-8



导致相电压增高, 励磁电流过大, 铁心饱和。
而铁心饱和后, 磁通量不再增加, 引起定子铜耗铁耗, 可能引起烧毁。铁心饱和程度增加, 励磁电流有增大增加。
负载运行时定子电流会大于额定电流, 而铁心饱和程度增加。

5-13

$U_{1N} = 380V$, $f_N = 50Hz$, $P_N = 7.5kW$, $n_N = 960r/min$

定子绕组为三角形联接, $2P=6$, $\cos \phi_N = 0.82$, $P_{Cu1} = 470W$, $P_{Fe} = 234W$, $P_{mech} = 40W$, $P_0 = 80W$

求满载时的: (1) 转差率 (2) 转子电流频率 (3) 转子铜耗 (4) 定子电流 (5) 定子铜耗 (6) 定子铁耗 (7) 定子效率 (8) 定子功率因数 (9) 定子功率因数角 (10) 定子功率因数角正切 (11) 定子功率因数角余切 (12) 定子功率因数角正切平方 (13) 定子功率因数角余切平方 (14) 定子功率因数角正切平方加一 (15) 定子功率因数角余切平方加一

$$f_2 = f_1 \cdot s = 50 - 3.8\% \times 50 = 48.1Hz$$

$$P_2 = P_N + P_{Cu2} + P_{mech} + P_0$$

$$P_{mech} = P_N + P_{Cu2} + P_{mech} + P_0$$

$$P_{Cu2} = P_N + P_{Cu2} + P_{mech} + P_0$$

$$P_{Cu2} = \frac{P_N + P_{mech}}{s - 1} = \frac{7.5kW + 40W}{0.038 - 1} = 298W$$

$$P_1 = P_2 + P_0 + P_{Fe} + P_{Cu1} = 7842W + 80W + 234W + 470W = 8626.1W$$

$$\eta = \frac{P_2}{P_1} = 87\%$$

$$I_1 = \frac{P_1}{\sqrt{3} U_1 \cos \phi_1} = 9.14A$$

$$I_1 = \frac{P_1}{\sqrt{3} U_1 \cos \phi_1}$$

5-15 三相异步电动机 $P_N = 10 \text{ kW}$, $U_N = 380 \text{ V}$, $I_N = 11 \text{ A}$.

定子绕组为Y联结, 额定运行时, $P_{cu} = 357 \text{ W}$, $P_{cu2} = 314 \text{ W}$, $P_{Fe} = 276 \text{ W}$, $P_{mech} = 71 \text{ W}$, $P_{200 \text{ W}}$

额定转速

空载转速

电动机转速

电动机机械输出功率

④ $P_N = 10 \text{ kW}$, $P_2 = P_N + P_{cu} + P_{cu2} + P_{Fe} = 10 \text{ kW} + 314 \text{ W} + 71 \text{ W} + 200 \text{ W} = 10591 \text{ W}$

⑤ $P_{cu2} = s P_2$, $s = 0.03$

$n_s = \frac{60 \times 10}{2} = 1500 \text{ r/min}$, $n = (1-s)n_s = 1455 \text{ r/min}$

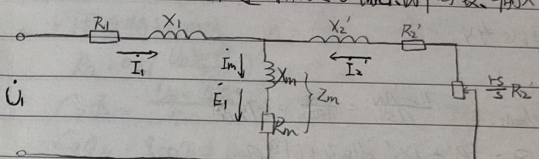
② $T_0 = \frac{P_0}{\omega} = \frac{P_{mech} + P_{200 \text{ W}}}{\omega} = \frac{71 \text{ W} + 200 \text{ W}}{2\pi \times 1455 / 60} = 1.82 \text{ N}\cdot\text{m}$

③ $T_e = \frac{P_2}{\omega} = \frac{P_N}{\omega} = \frac{10000 \text{ W}}{2\pi \times 1455 / 60} = 65.7 \text{ N}\cdot\text{m}$

④ $T_2 = \frac{P_2}{\omega} = \frac{P_N}{\omega} = 65.7 \text{ N}\cdot\text{m}$

⑤ $T_2 = \frac{P_2}{\omega} = \frac{P_N}{\omega} = 65.7 \text{ N}\cdot\text{m}$

5-14 $P_N = 10 \text{ kW}$, $U_N = 380 \text{ V}$, $f_N = 50 \text{ Hz}$, $n_N = 1455 \text{ r/min}$, $R_1 = 1.37 \Omega$, $X_1 = 2.43 \Omega$, $R_2' = 1.04 \Omega$, $X_2' = 4.4 \Omega$, $R_m = 8.34 \Omega$, $X_m = 82.6 \Omega$. 定子绕组为三角联结, 在额定负载时的机械损耗为 200 W . 求额定转速时的定子电流、功率因数、输入功率及效率。



$n_s = \frac{60 \times 10}{2} = 1500$, $s = \frac{n_s - n_N}{n_s} = \frac{1500 - 1455}{1500} = 3\%$, $\frac{1}{s} R_2' = \frac{1.04}{0.03} = 34.67 \Omega$

定子电流 $I_1 = \frac{U_1}{\sqrt{(R_1 + \frac{1}{s} R_2')^2 + (X_1 + X_2')^2}}$

① $jX_2' + \frac{1}{s} R_2' = 34.67 \Omega + j4.4 \Omega = 34.95 \angle 7.23^\circ \Omega$

② $Z_m = R_m + jX_m = 8.34 + j82.6 = 83 \angle 84.23^\circ \Omega$

③ $Z_1 = (26.44 + j13.93) \Omega = 29.9 \angle 27.78^\circ \Omega$

④ $Z = Z_1 + \frac{1}{1/s} = (27.815 + j16.36) \Omega = 32.2 \angle 30.46^\circ \Omega$

⑤ $U_1 = 380 \angle 0^\circ$, $I_1 = \frac{U_1}{Z} = \frac{380 \angle 0^\circ}{32.2 \angle 30.46^\circ} = 11.77 \angle -30.46^\circ \text{ A}$, $\varphi_1 = 30.46^\circ$

$P_1 = 3 U_1 I_1 \cos \varphi_1 = 3 \times 380 \times 11.77 \times \cos 30.46^\circ = 11539 \text{ W}$

$\eta = \frac{P_2}{P_1} = \frac{10 \text{ kW}}{11539 \text{ W}} = 86.6\%$

5-16 一台三相绕线转子异步电动机

$U_N = 380V, f_N = 50Hz$ 联结, $n_N = 1440r/min$

$R = 0.4\Omega, R_2' = 0.4\Omega, X_1 = 1\Omega, X_2' = 1\Omega, X_m = 40\Omega, R_m$ 略不计

转子为三相, 转子有效匝数为 4

4. 满载时的转差率

$$s = \frac{n_s - n}{n_s} = \frac{1500 - 1440}{1500} = 4\%$$

$$\frac{1}{s} R_2' = \frac{0.4}{0.04} = 10\Omega, Z = \frac{1}{s} R_2' + jX_2' = (10 + j1)\Omega$$

用近似等效电路计算

$$Z_m = R_m + jX_m = j40 = 40\angle 90^\circ \Omega$$

$$Z = Z_1 + Z_m // Z_2 = 11.28\angle 10.2^\circ \Omega$$

设 $U = 220\angle 0^\circ V, I_m = \frac{U}{Z_m} = 5.5\angle -90^\circ A$

$$I_2' = \frac{-0.220\angle 0^\circ}{11.28\angle 10.2^\circ} = -19.5\angle 70.2^\circ A$$

$$I_1 = I_m - I_2' = 21.18\angle -25^\circ A$$

$$(14) E_2' = I_2' (\frac{1}{s} R_2' + jX_2') = -209.6\angle -4.86^\circ V$$

$$E_2 = \frac{1}{k} E_2' = \frac{1}{4} \times 209.6V = 52.4V \quad E_2' = k E_2$$

匝数比

$$k_e = \frac{N_1 k_w1}{N_2 k_w2}$$

$$k_i = \frac{m_1}{m_2} \cdot \frac{N_1}{N_2} \times \frac{k_w1}{k_w2}$$

$$E_{2s} = s E_2 = 0.04 \times 52.4V = 2.1V$$

$$f_2 = s f_1 = 1.865Hz$$

4. 总机械功率 $P_{mech} = 3I_2'^2 \frac{1-s}{s} R_2' = 81172W$

1) 额定电磁转矩 $T_e = \frac{P_{mech}}{\omega_2} = \frac{81172}{2\pi \times 1440/60} = 170N \cdot m$

9-1, 9-2, 9-3, 9-14

9-1. 他励电动机的铭牌数据: $P_N = 1.75kW, U_N = 110V, I_N = 20A, n_N = 1440r/min$
试计算: <<他励>>

1. 固有特性曲线 $n = \frac{U_N}{C_e \Phi} - \frac{R_a}{C_e \Phi} I_a$

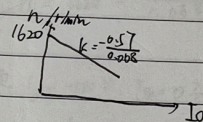
$$R_a = 0.5 (\frac{U_N I_N}{P_N}) = 0.57\Omega$$

$$R_a = (\frac{1}{2} - \frac{3}{5}) \frac{U_N I_N - P_N}{I_N^2}$$

$$C_e \Phi = \frac{U_N - I_N R_a}{I_N} = 0.068 \quad n_0 = \frac{U_N}{C_e \Phi} = 1620r/min$$

$$C_T \Phi = 9.55 C_e \Phi I_N = 9.55 \times 0.068 = 0.65$$

$$n = 1620 - \frac{0.57}{0.068} I_a$$



2. $I_a = \frac{1}{2} \times 20A = 10.05A, n = 1620 - \frac{0.57}{0.068} \times 10.05 = 1536r/min$

3. $n = 1500r/min$ 时, 由机械特性方程 $I_a = \frac{1620 - 1500}{8.38} = 14.32A$

9-2 他励直流电机的铭牌数据同上, 试计算磁通为 80% 额定电压为 50% U_N 时的人为特性, 并在坐标纸上画出。

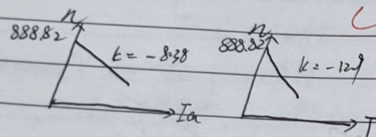
$$n = \frac{U_N - I_a R_a}{C_e \Phi} = 2022 - 10.48 I_a$$

$$* = \frac{U_N - I_a R_a}{C_e \Phi} = 2022 - 20.15 I_a$$

50% U_N : $U = 51V$

$$n = \frac{U - I_a R_a}{C_e \Phi} = \frac{51}{0.068} - 8.38 I_a = 808.82 - 8.38 I_a$$

$$n = \frac{U - I_a R_a}{C_e \Phi} = \frac{51}{0.068} - 12.9 I_a$$



9-3 题 9-1 中的电动机用于起吊和下放重物

1. 当额定负载时, 电枢电路中有 50% R_N 或 50% R_N ($R_N = U_N / I_N$) 时电动机稳定运转转速为多少? 各处于何种运转状态? 只写分数, 1, 2, 3, 4, ...

$$R_N = \frac{110}{20.1} \Omega = 5.47 \Omega$$

40% $R_N = 0.5 R_N$ 时

$$n = 1620 - \frac{5.47 \times 0.5 \times 20.1}{0.068} \times 20.1 = 643 \text{ r/min}$$

电动运行 吊起重物

60% $R_N = 1.5 R_N$ 时

$$n = 1620 - \frac{5.47 \times 1.5 \times 20.1}{0.068} \times 20.1 = -974 \text{ r/min}$$

反接制动 下放重物

2. $U = -110V$, $I_a = 0.3 I_N = 0.3 \times 20.1A = 6.03A$.

$$n = (-1620 - \frac{6.03 \times 5.47}{0.068}) \text{ r/min} = -1671 \text{ r/min}$$

回馈制动 下放重物

3. $U = 0$, $R_a = 0.5 R_N = 0.5 \times 5.47 \Omega = 2.735 \Omega$.

$$n = -\frac{2.735 + 5.47}{0.068} \times 20.1 \text{ r/min} = -977 \text{ r/min}$$

能耗制动 下放重物

个力后

9-14. $P_N = 18.5 \text{ kW}$, $U_N = 220 \text{ V}$, $I_N = 103 \text{ A}$, $n_N = 1500 \text{ r/min}$, $n_{\text{max}} = 1500 \text{ r/min}$, $R_A = 0.18 \Omega$

$$\text{① } C_{\Phi} = \frac{1}{2} C_{\Phi N} = \frac{1}{2} \times \frac{220 - 103 \times 0.18}{220} = 0.1343$$

② 电机电流增加到原来的3倍

$$I_a = 3 \times 103 \text{ A} = 309 \text{ A}, \quad n = \frac{220 - 309 \times 0.18}{0.1343} \text{ r/min} = 1227 \text{ r/min}$$

③ 带恒功率负载，磁通减弱为额定值的 $\frac{1}{3}$ ，由 $P_2 = P_N$ 得

$$\frac{I_{a\text{max}}}{I_{a\text{N}}} = \frac{T_{\text{max}}}{T_{\text{N}}}$$

$$T = T_{\text{N}} \frac{P_{\text{N}}}{P_{\text{max}}} = T_{\text{N}} \frac{18.5}{55.5} = \frac{1}{3} T_{\text{N}}$$

$$T = \frac{1}{2} C_{\Phi} I_a n = C_{\Phi} I_N n$$

$$\frac{1}{3} C_{\Phi} I_a n = \frac{1}{2} C_{\Phi} I_N n, \quad I_a = I_N = 103 \text{ A}$$

$$n = \frac{220 - 103 \times 0.18}{0.1343} \text{ r/min} = 1500 \text{ r/min}$$

电动机能长期运行，电机电流为电动机的额定电流

阅, 2015, 12, 18